Trader Behavior vs Market Sentiment Analysis

Methodology

Data Sources. The analysis combines two datasets: (1) a Fear & Greed sentiment dataset containing 2,644 observations and 4 features (Timestamp, Fear & Greed Value, Sentiment Classification, and Date), and (2) a historical trading dataset containing 211,224 trades and 16 features, including Account, Coin, Execution Price, Size USD, Size Tokens, Direction, Closed PnL, Fee, and Timestamp.

Data Preparation. Data quality checks confirmed that both datasets contained no missing values or duplicate records. Timestamps were converted into a common datetime format and aligned at the daily level. The datasets were then merged using the trading date. Several analytical features were engineered, including Daily PnL, Win Rate, Average Trade Size, Drawdown Proxy, Long/Short Ratio, Trade Frequency, and a Position Aggressiveness Proxy.

Analysis. The study evaluates trader performance across different sentiment regimes using profitability, win rate, drawdown, and statistical significance testing. Behavioral analysis examines trade frequency, position sizing, long/short bias, and aggressiveness. Traders were further segmented into High vs Low Aggressiveness, Frequent vs Infrequent Traders, and Consistent Winners vs Inconsistent Traders to identify performance differences across trading styles.

Key Insights

1. Market sentiment significantly affects profitability

Average profitability during Greed periods was **87.89** compared with **50.05** during Fear periods, representing an increase of approximately **76%**. Win rates also improved from **41.5%** during Fear periods to **44.6%** during Greed periods. A Welch t-test confirmed that the difference is statistically significant ($p < 0.05$).

2. Fear periods are associated with higher risk and activity

Fear-related periods exhibited larger drawdowns and substantially higher trading activity than Greed periods. This suggests that traders react more aggressively during uncertain market conditions, resulting in greater downside risk.

3. Conservative traders outperform aggressive traders

Trader segmentation revealed that lower-aggressiveness traders generated significantly higher average profitability (**73.65**) than highly aggressive traders (**27.29**). Similarly, infrequent traders outperformed frequent traders, indicating that trade quality is more important than trade quantity.

Strategy Recommendations

Strategy 1: Reduce Position Aggressiveness During Fear Periods

Fear periods were associated with lower profitability and larger drawdowns. Traders should reduce position aggressiveness and prioritize capital preservation when market sentiment deteriorates.

Strategy 2: Prioritize High-Conviction Trades

Infrequent traders achieved substantially higher profitability than frequent traders. Rather than increasing trading frequency, traders should focus on selective, higher-quality opportunities with favorable risk-reward characteristics.

Predictive Model

A Random Forest classifier was trained using Fear & Greed values, trade size, and position aggressiveness features. The model achieved approximately **68% accuracy**, suggesting that sentiment and behavioral variables contain useful predictive information regarding future trade profitability.